

Problem 290: Character Counter

Difficulty: Medium

Author: Adrienne McKee, Huntsville, Alabama, United States

Originally Published: Code Quest 2026

Problem Background

Sometimes we want to send messages to people but are limited on the amount of space to send that message. For example, X (formerly known as Twitter), engravings on jewelry, and the first generation of SMS text messages have all had limits on the amounts of characters that can be used. You want to write a program that calculates the length, number of words, and number of times each character is used in a particular message.

Problem Description

Your program should read in a line of characters and calculate the total length of the text, the number of words in the text (separated by spaces), and the number of times each character is used. The program will then display the message again, the number of words and characters it contains, as well as all the characters that are used in the message and the total number of times each of them are used. The list of characters should display the most used character first and proceed downward from there.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include a single line containing characters to be analyzed. "Words" are defined as a series of characters separated by a space.

2

Code Quest Is The Best
Yabba-dabba-doo!

Sample Output

For each test case, your program must print:

- A line with the input characters being analyzed.
- A line with the hyphen (-) character, repeated once for each character in the message (including spaces).
- A line with the word "CHARACTERS" followed by a colon, a space, and the number of characters (including spaces).

- A line with the word “WORDS” followed by a colon, a space, and the number of words.
- One line for each unique character appearing in the message, containing the character, a colon, a space, and the number of times that character appears in the message.
 - Characters should be sorted in decreasing order by number of occurrences.
 - If two or more characters appear the same number of times, they should be ordered according to their first appearance in the message (first character occurrence appears first amongst the tied characters).

Code Quest Is The Best

CHARACTERS: 22

WORDS: 5

e: 4

: 4

s: 3

t: 2

C: 1

o: 1

d: 1

Q: 1

u: 1

I: 1

T: 1

h: 1

B: 1

Yabba-dabba-doo!

CHARACTERS: 16

WORDS: 1

a: 4

b: 4

-: 2

d: 2

o: 2

Y: 1

!: 1